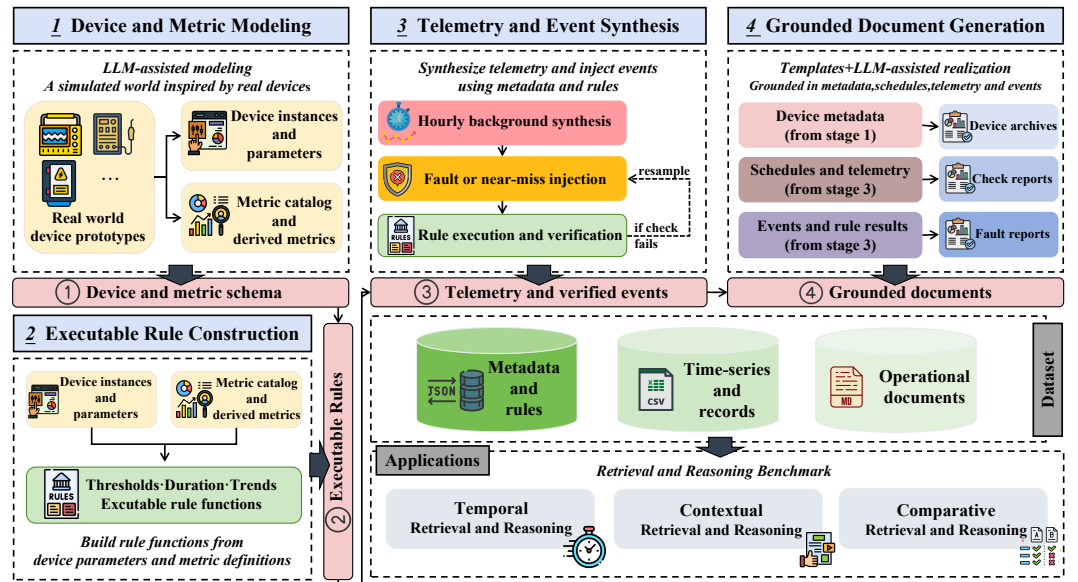

1 Device and Metric Modeling
LLM-assisted modeling
A simulated world inspired by real devices
Real world device prototypes
Device instances and parameters
Metric catalog and derived metrics
3 Telemetry and Event Synthesis
Synthesize telemetry and inject events using metadata and rules
Hourly background synthesis
Fault or near-miss injection
resample
Rule execution and verification
if check fails
4 Grounded Document Generation
Templates+LLM-assisted realization
Grounded in metadata,schedules,telemetry and events
Device metadata (from stage 1)
Device archives
Schedules and telemetry (from stage 3)
Check reports
Events and rule results (from stage 3)
Fault reports
① Device and metric schema
③ Telemetry and verified events
④ Grounded documents
2 Executable Rule Construction
Device instances and parameters
Metric catalog and derived metrics
Thresholds·Duration·Trends
Excutable rule functions
Build rule functions from device parameters and metric definitions
② Executable Rules
Metadata and rules
Time-series and records
Operational documents
Dataset
Applications
Retrieval and Reasoning Benchmark
Temporal
Retrieval and Reasoning
Contextual
Retrieval and Reasoning
Comparative
Retrieval and Reasoning